Image Classification on the Caltech-101 Dataset: A Comparison of Classical Machine Learning and Deep Learning Methods

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1. Introduction

Image classification is a fundamental task in computer vision, aiming to assign semantic labels to images based on their visual content. Over the past decade, deep learning methods, particularly convolutional neural networks (CNNs), have significantly advanced image recognition by learning hierarchical feature representations directly from raw pixel data. However, classical machine learning approaches based on hand-crafted features remain valuable baselines and provide insight into the advantages of modern deep learning techniques.

The goal of this project is to compare classical and deep learning methods for multi-class image classification using the Caltech-101 dataset. Specifically, we evaluate a traditional feature-based pipeline using Histogram of Oriented Gradients (HOG) with a Support Vector Machine (SVM), as well as two deep learning models based on transfer learning: ResNet-18 and EfficientNet-B0. These models were selected to represent different levels of model complexity and architectural design strategies.

We systematically evaluate model performance using multiple metrics, including overall accuracy, per-class accuracy, confusion matrices, and F1-scores. In addition, we conduct ablation studies to analyze the impact of input resolution and data augmentation on deep learning performance. Through these experiments, we aim to understand the relative strengths of classical and modern approaches and identify factors that most strongly influence classification accuracy.

2. Dataset

The experiments in this project were conducted using the Caltech-101 dataset, which contains images from 101 object categories along with a background category. The dataset includes approximately 9,000 images with significant variation in object appearance, pose, scale, and background clutter. The number of samples per category is imbalanced, typically ranging from 40 to 800 images per class, which presents additional challenges for classification models.

Images were obtained from a publicly available Kaggle version of the dataset. All images were resized to a fixed resolution depending on the experiment (either 64 × 64 or 128 × 128 pixels) and normalized prior to training.

To ensure fair evaluation, the dataset was split into training, validation, and test sets using a stratified 70/15/15 ratio, preserving the class distribution across splits. The validation set was

used for model selection and hyperparameter tuning, while the test set was reserved exclusively for final performance evaluation.

The dataset is publicly available at: https://www.kaggle.com/datasets/imbikramsaha/caltech-101

3. Methods

3.1 Classical Machine Learning Baseline

As a classical baseline, we implemented an image classification pipeline based on Histogram of Oriented Gradients (HOG) features combined with a linear Support Vector Machine (SVM) classifier. All images were resized to 64 × 64 pixels and converted to grayscale prior to feature extraction. HOG descriptors were computed using 9 orientation bins, 8 × 8 pixel cells, and 2 × 2 block normalization.

The dataset was split into training, validation, and test sets using a stratified 70/15/15 ratio to preserve class distribution across splits. A LinearSVC model with regularization parameter C=1.0 was trained on the extracted HOG features.

3.2 Deep Learning Models

To improve performance beyond classical hand-crafted features, we implemented two convolutional neural network models using transfer learning with ImageNet-pretrained weights.

ResNet-18

The ResNet-18 architecture was initialized with ImageNet weights, and the final fully connected layer was replaced to match the 101 output classes of the Caltech-101 dataset. Images were resized to 128 × 128 pixels. Data augmentation techniques including random horizontal flipping and color jitter were applied during training to improve generalization.

The model was trained for 5 epochs using the Adam optimizer with a learning rate of 1e-4 and cross-entropy loss.

EfficientNet-B0

We also evaluated EfficientNet-B0, which introduces a compound scaling strategy that jointly scales network depth, width, and resolution. Similar to ResNet, the final classification layer was replaced to match the dataset classes, and training used the same preprocessing, optimizer, and loss configuration.

3.3 Evaluation Metrics

Each model was evaluated using multiple metrics:

- Overall accuracy

- Per-class accuracy
- Confusion matrix
- Precision, recall, and F1-score (macro and weighted averages)

These metrics provide complementary insights into model performance, particularly under class imbalance conditions present in the Caltech-101 dataset.

4. Results

4.1 Classical Baseline Performance

The HOG + SVM model achieved the following performance:

- Validation Accuracy: 0.61
- Test Accuracy: 0.595
- Macro F1-score: 0.407
- Weighted F1-score: 0.588

Performance varied significantly across categories. Classes with distinctive structural features such as Motorbikes, Faces, and Airplanes achieved high accuracy, while visually complex or small-sample classes performed poorly. Some categories received zero precision because the classifier failed to predict any samples for those classes.

Figure 1 shows the confusion matrix of predictions on the test set, and Figure 2 presents the per-class accuracy for the worst-performing categories.

4.2 ResNet-18 Performance

The ResNet-18 model achieved substantially higher performance compared to the classical baseline:

- Validation Accuracy: 0.93
- Test Accuracy: 0.933
- Macro F1-score: 0.904
- Weighted F1-score: 0.930

Figure 3 shows the normalized confusion matrix of the model predictions. Compared with the classical method, the matrix exhibits a much stronger diagonal structure, indicating more consistent classification across categories. Figure 4 presents the per-class accuracy for the lowest-performing categories.

4.3 EfficientNet-B0 Performance

The EfficientNet-B0 model achieved the best performance among all evaluated methods:

- Test Accuracy: 0.946

- Macro F1-score: 0.919
- Weighted F1-score: 0.944

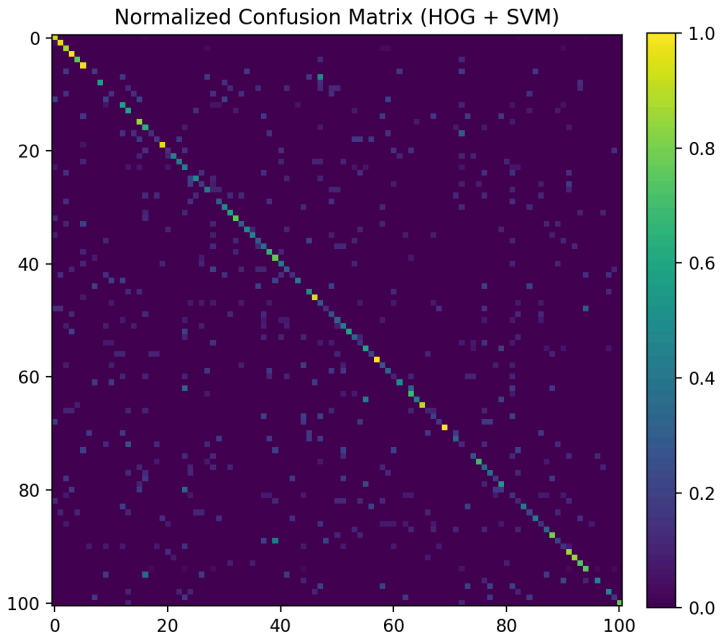

Figure 1: Normalized confusion matrix for HOG + Linear SVM on the Caltech-101 test set.

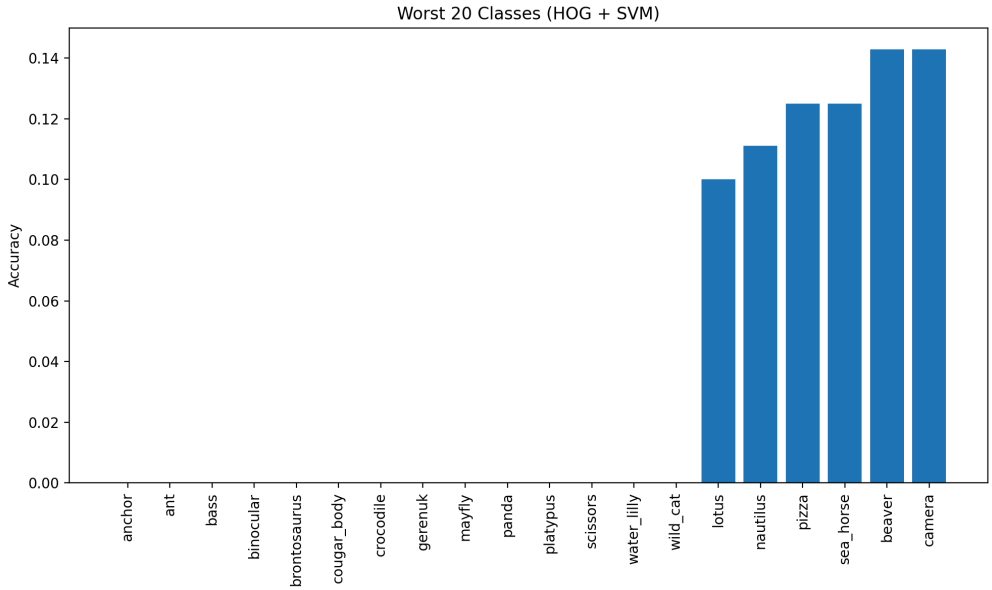

Figure 2: Per-class accuracy for the worst 20 categories using the HOG + SVM model.

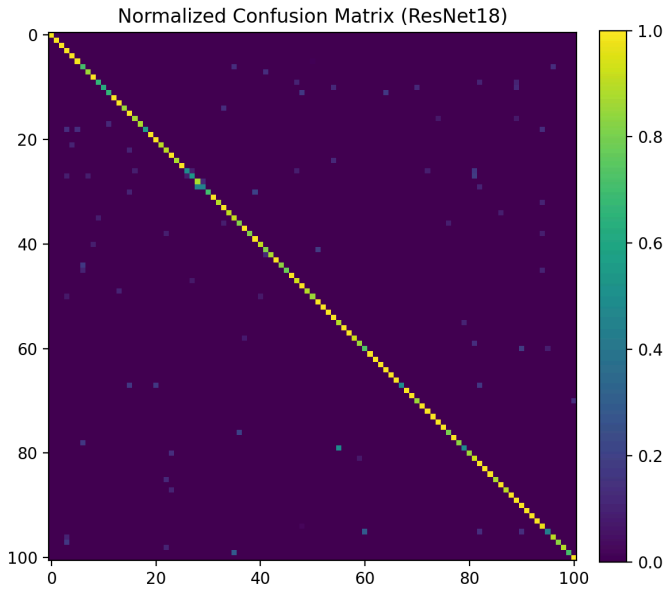

Figure 3: Normalized confusion matrix for the ResNet-18 model on the Caltech-101 test set.

The training and validation loss curves (Figure 5) show stable convergence with minimal overfitting. Figure 6 presents the normalized confusion matrix, demonstrating strong diagonal dominance across most categories. Figure 7 illustrates the per-class accuracy for the most challenging classes, which remain substantially higher than those observed in the classical baseline.

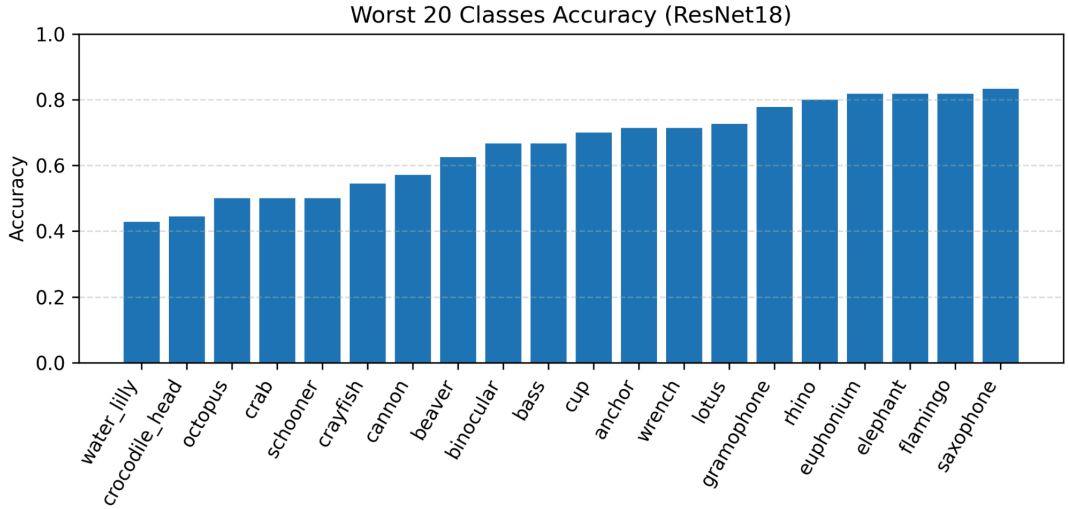

Figure 4: Per-class accuracy for the 20 lowest-performing categories using the ResNet-18 model.

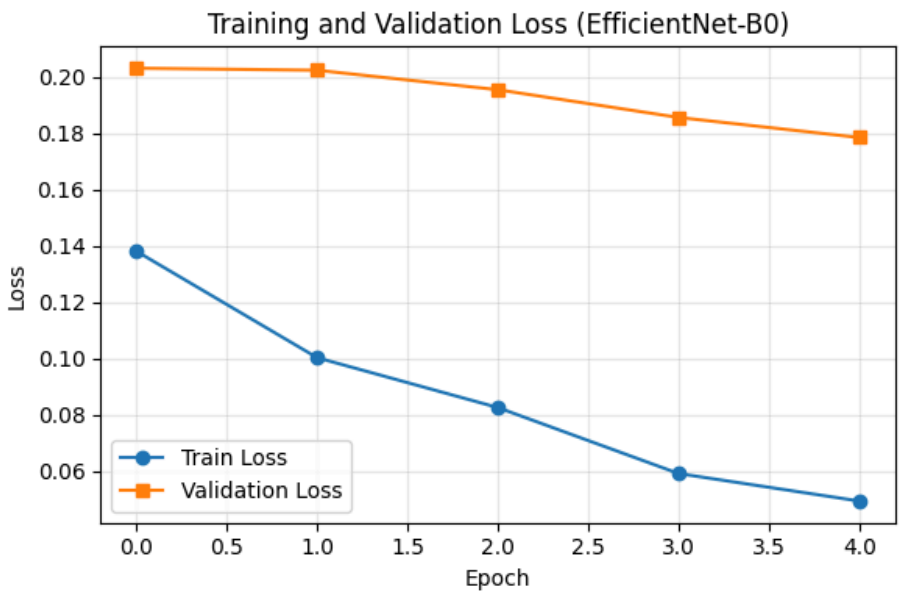

Figure 5: Training and validation loss curves for the EfficientNet-B0 model.

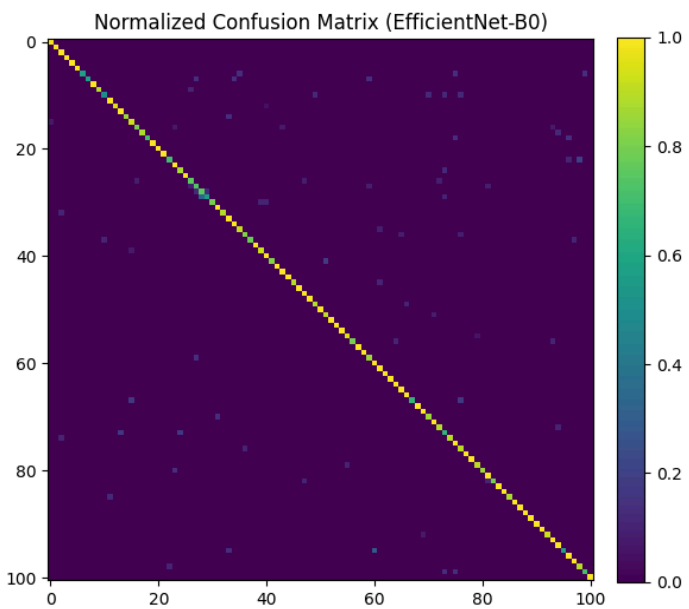

Figure 6: Normalized confusion matrix for EfficientNet-B0 predictions on the test set.

5. Discussion

The experimental results demonstrate a clear performance hierarchy across the evaluated approaches. The classical HOG-based method achieved moderate accuracy but struggled with fine-grained object distinctions and categories with limited training data. This limitation comes from the inability of hand-crafted features to capture complex semantic information and intra-class variability.

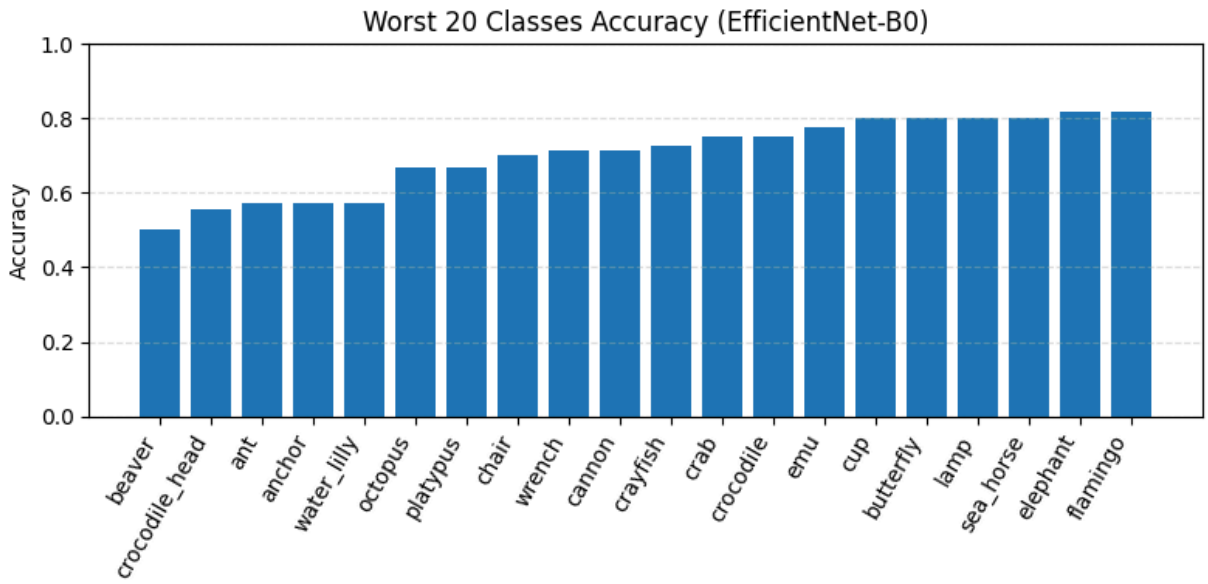

Figure 7: Per-class accuracy for the 20 lowest-performing categories using EfficientNet-B0.

In contrast, both deep learning models significantly improved performance through learned hierarchical representations. Transfer learning with pretrained networks proved highly effective, enabling strong generalization even with a relatively limited number of training samples per category.

EfficientNet-B0 achieved the highest accuracy among all models, likely due to its compound scaling strategy, which allows more efficient feature extraction and improved representation capacity compared to ResNet-18. Remaining errors were primarily associated with visually similar categories or classes with few samples.

Overall, the results confirm that modern convolutional neural networks substantially outperform classical feature-based methods for multi-class image classification tasks such as Caltech-101.

6. Ablation Study

6.1 Effect of Image Resolution

To evaluate the impact of input resolution, we compared image sizes of 128 × 128 and 64 × 64 using the EfficientNet-B0 model while keeping all other training settings constant.

Reducing the input resolution from 128 × 128 to 64 × 64 resulted in a significant performance drop, with test accuracy decreasing from 94.6% to 75.3%. This large reduction suggests that higher-resolution images provide critical spatial information necessary for accurate classification in the Caltech-101 dataset.

The performance degradation was particularly pronounced for fine-grained categories and classes with limited training samples, where detailed texture and shape information play an

important role. These results demonstrate that input resolution is a key factor influencing deep learning performance in multi-class image classification tasks.

6.2 Effect of Data Augmentation

To evaluate the impact of data augmentation, we trained the EfficientNet-B0 model with and without augmentation while keeping the input resolution (128 × 128), optimizer settings, and training procedure identical. The augmentation strategy included random horizontal flipping, small rotations, and color jitter.

Removing data augmentation resulted in nearly identical performance compared to the augmented model, with test accuracy changing from 94.6% to 94.5%. This small difference suggests that augmentation provided limited additional benefit in this task. One possible explanation is that the model was initialized with pretrained ImageNet weights, which already capture robust visual features and improve generalization. Additionally, the Caltech-101 dataset contains sufficient variation across samples, reducing the reliance on synthetic transformations.

Table 1: Performance comparison for EfficientNet-B0 under different input resolutions and data augmentation settings.

Model	**Image Size**	**Augmentation**	**Test Accuracy**
EfficientNet-B0	128×128	Yes	0.9455
EfficientNet-B0	128×128	No	0.9448
EfficientNet-B0	64×64	No	0.7527

7. Conclusion

This project compared classical and deep learning approaches for multi-class image classification on the Caltech-101 dataset. The classical HOG + SVM baseline achieved moderate performance but struggled with complex visual patterns and class imbalance. Deep learning models using transfer learning significantly improved accuracy, with ResNet-18 achieving over 93% accuracy and EfficientNet-B0 achieving the best performance at approximately 94.6%.

Ablation experiments demonstrated that input resolution plays a critical role in model performance, while data augmentation had limited impact when using strong pretrained architectures. Overall, the results highlight the effectiveness of transfer learning and modern convolutional neural networks for image classification tasks with limited training data.

8. Code Availability

The code, implementation and experiments are publicly available at:
https://github.com/xiaoman21/caltech101-project

References

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